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|  | Document Type:  | <b>DOCUMENTED INFORMATION</b><br>ISO 9001:2015 | Document Code  | <b>RES- PROP-01</b>      |
|                                                                                   | Document Title: | <b>RESEARCH PROPOSAL</b>                       | Revision No.   | <b>00</b>                |
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|                                                                                                                                                    |                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>(1) Research Program/Project/Study Title:</b><br><b>GERMINATION, GROWTH AND YIELD, AND POSTHARVEST QUALITY OF ANTHILL SOIL-GROWN VEGETABLES</b> |                                                                                                                                                                                                                                                          |
| <b>Proponent Name:</b>                                                                                                                             | <b>DIEGO M. MALONES</b>                                                                                                                                                                                                                                  |
| <b>Academic Rank:</b>                                                                                                                              | <b>ASSOCIATE PROFESSOR V</b>                                                                                                                                                                                                                             |
| <b>Co-Authors (with academic rank):</b>                                                                                                            | <b>JON VERNY D. BIACO (INSTRUCTOR 1), EDEN A. ALAM (INSTRUCTOR 1), JUNAREY B. DOLETIN (INSTRUCTOR 1), JASPER A. BASALO (INSTRUCTOR 1), HAROLD O. BUENVENIDA (ASST. PROF IV), DAREL D. DAPULAZA (INSTRUCTOR 1), and MARTIN O. TON-OGAN (INSTRUCTOR 1)</b> |
| <b>Campus/Satellite College/Address:</b>                                                                                                           | <b>CAPSU PONTEVEDRA</b>                                                                                                                                                                                                                                  |
| <b>Telephone/Cell Phone/Fax Number:</b>                                                                                                            | <b>09283882295</b>                                                                                                                                                                                                                                       |
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

|                            |  |
|----------------------------|--|
| <b>Component Title:</b>    |  |
| <b>Component Leader/s:</b> |  |
| <b>Academic Rank:</b>      |  |

|                                                                                                                                                                                                                                                                                                                                                          |  |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>(2) Category (please check)</b><br><input type="checkbox"/> Social Science/Education/Humanities & other related fields<br><input checked="" type="checkbox"/> Natural/Biological Sciences<br><input type="checkbox"/> Food Research<br><input type="checkbox"/> Developmental/Computer/Engineering/Architecture<br><input type="checkbox"/> Extension |  | <b>(3) Research Center</b><br>[Check the concerned center]<br><br><input type="checkbox"/> SSITRDC<br>Social Science and Industrial Technology Research Development Center<br><br><input type="checkbox"/> LRDC<br>Livestock Research Development Center<br><br><input type="checkbox"/> FFRDC<br>Fisheries and Food Research Development Center<br><br><input checked="" type="checkbox"/> CSRDC<br>Crop Science Research Development Center |
| <b>(4) Site of implementation</b>                                                                                                                                                                                                                                                                                                                        |  |                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Municipality/City:                                                                                                                                                                                                                                                                                                                                       |  | <b>Pontevedra</b>                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Province:                                                                                                                                                                                                                                                                                                                                                |  | <b>Capiz</b>                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Region:                                                                                                                                                                                                                                                                                                                                                  |  | <b>VI</b>                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>(5) Classification</b><br><br><b>Research:</b> <b>Development:</b><br><input checked="" type="checkbox"/> Basic <input type="checkbox"/> Pilot Testing<br><input type="checkbox"/> Applied <input type="checkbox"/> Tech. Promotion/Commercialization                                                                                                 |  | <b>(6) Mode of Implementation</b><br><br><input type="checkbox"/> Single Authorship/Proponent<br><input type="checkbox"/> Collaborative Research (within campus)<br><input checked="" type="checkbox"/> Collaborative Research (intercampus)                                                                                                                                                                                                  |

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| <b>(7) Sector/Commodity</b><br><br><b>CROP SCIENCE</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  | <b>(8) Discipline/Field of Specialization</b><br><b>CROP</b>      |
| <p><b>(9) Rationale</b>(State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R &amp; D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)</p> <p>Vegetable production is essential for providing food security and eradicating poverty. In response to the COVID-19 outbreak, families are showing an increased interest in growing their own veggies for consumption. The traditional soil media, which consists of garden soil, sand, and animal manure, is used by homes to grow vegetables in pots in places with a limited amount of space. In addition to being a significant new cash crop for agriculture, vegetables are a great source of readily available vitamins and minerals that may strengthen people's immune systems against infectious diseases like the COVID-19 virus.</p> <p>To meet the crops' nutrient needs, inorganic fertilizer is currently employed heavily in the cultivation of vegetables. This method of growing veggies is used by growers who raise their produce in pots as well. Because of this method's long-standing use, the soil has become less friable, more acidic, and has a lower level of organic matter, all of which make it more difficult for plants to absorb the nutrients they need for better growth and development. The possibility of this technique succeeding is nevertheless low due to problems with smallholder farmers' affordability and accessibility.</p> <p>However, it's time to think about a different approach that farmers with limited resources might use to increase agricultural productivity without paying exorbitant costs. According to a review by Chisanga et al. (2020), anthills have supposedly been employed as a low-external-input agricultural strategy to alleviate soil fertility difficulties in many African countries. Farmers in southern Zambia dig and gather soil from anthills and utilize it as an alternative to increase soil fertility at the farm level. Anthill and termite mound soils have been given to farmers in Nigeria, Zimbabwe, and Uganda in recent years (Chisanga et al., 2017).</p> <p>Smallholder farmers who grow vegetables and grains in the Philippines do not frequently use anthill soil. It is important to look into whether anthill or termite mound soils may replace synthetic fertilizers with less expensive, more sustainable alternatives. Nearly all of the country's agro-ecological regions have a plentiful supply of anthills or termite mounds. However, very little to no research has been conducted up to this point to evaluate the use of anthill and termite soils as sources of nutrients for the production of popular vegetables in the country.</p> |  |                                                                   |
| <p><b>(9-A) Project Description</b> (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |                                                                   |
| <p><b>(10) Objectives/Statement of the Problem</b> (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)</p> <p>The aim of this study is to analyze the physicochemical properties of anthill soils in combination with a standard soil medium and assess the germination, growth, yield, and postharvest quality of common vegetables such as pechay, lettuce, and mustard. The objectives are as follows:</p> <p>(1) to examine the physicochemical properties of various anthill soil and standard soil medium combinations;</p> <p>(2) to assess the seed germination of pechay, lettuce, and mustard in response to various anthill soil and standard soil medium combinations;</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |                                                                   |

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- (3) to assess the growth and yield response of pechay, lettuce, and mustard to various anthill soil and standard soil medium combinations;
- (4) to assess the quality of harvested pechay, lettuce, and mustard; and
- (5) to analyze the cost and return of producing vegetables grown in various anthill soil and standard soil medium combinations.

**(11) Scientific Basis/Theoretical Framework**(Include concepts, theories and variables that indicate relevant relationships to be tested.

**(12) Review of Literature/Prior Art Search**(Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

### **Anthill: Its Nature and Importance**

Many academics believe that termites and other soil-dwelling fauna species have an impact on the development of anthills. Ants that create anthills collect woody debris for their nests and go on extensive insect hunts in search of honeydew and other food sources for their colony. Apparently, compared to neighboring soils, active anthills contain higher soil organic matter and inorganic nutrients like Ca, K, Mg, Na, and P. (Folgarait, 1998; Kristiansen et al., 2001). According to Wang et al. (1995), ant activities considerably change the physical properties of the soil, such as infiltration and porosity; the faunal biomass and microbial community of the soil; and the rates of organic matter decomposition (Petal and Kuisinaka, 1994). The actions of ground-dwelling ants are apparent when anthills are being constructed. Ant construction creates nutrient-rich niches in the soil below that encourage seed sprouting (Levey and Byrne, 1993; Andersen and Morrison, 1998). Other studies have linked these behaviors to plant distribution patterns (Dean et al., 1997; Garrettson et al., 1998) and vegetation succession (Farji-Brener and Silva, 1995). Anthill construction by ants has been linked by scientists to changes in soil's physical-chemical properties (Nkem et al., 2000; Lenoir et al., 2001; Lafleur et al., 2002).

Anthill soils are known for their reputation for having a generally thicker texture and a stronger capacity to retain water. Additionally, they possess high calcium and other base values (Phimister, 1974). Anthill soils are a fantastic natural source of nutrients since it seems like ants recycle nutrients in them. Anthill soils are believed to prevent nutrient losses and serve as a kind of manure, preserving soil moisture and texture (Africa Farm News, 2014). Anthill soil utilization includes the stages of digging, piling, and distributing soil on the field. Because it contains high quantities of organic carbon, total phosphorus, and bioavailable nitrogen, anthill soil is used as crop fertilizer by African farmers, according to Lopez-Hernandez et al. (2001). Scholars in Zambia (Siame, 2005), Uganda (Okwakol and Sekamatte, 2007), Zimbabwe (Bellon et al., 1999; Nyamapfene, 1986), Sierra Leone (Ettema, 1994), and Niger have reported on farmers' usage of anthill soil in agricultural productivity (Brouwer et al., 1993). Researchers have documented the use of anthill soil by farmers in the agricultural productivity of Zambia (Siame, 2005), Uganda (Okwakol and Sekamatte, 2007), Zimbabwe (Bellon et al., 1999; Nyamapfene, 1986), Sierra Leone (Ettema, 1994), and Niger (Brouwer et al., 1993). Farmers either plant specific crops on anthills or spread anthill dirt over their fields, according to Nyamapfene (1986) and Logan (1992). According to reports, Malawian farmers have planted a variety of crops adjacent to anthills, including bananas (*Musa* spp.). In Uganda, however, farmers cultivate anthills in addition to maize, tomatoes, onions, pumpkins, and other cucurbita species (Okwakol and Sekamatte, 2007).

In Zimbabwe, late-season planted maize, sweet sorghum (*Sorghum* spp.), okra (*Abelmoschus esculentus*), and pumpkins are practically cultivated on anthills and all require adequate water and nutrient delivery (Nyamapfene, 1986). Smallholder farmers in Niger reportedly prefer to grow sorghum atop anthills to the adjacent soils, according to Brouwer et al. (1993). Farmers in some regions clear anthills from their crops before spreading the soil. For instance, farmers in southern Zambia remove a portion of the anthill while taking care to protect the base and colony. This soil is then brought to the field and mixed with the top soil before it begins to rain.

In areas where conservation farming is practiced, planting basins and ripped lines are filled with anthill soil (Siame, 2005). In South Africa, sugarcane fields grown on sandy soils are marked by some

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remarkable, well-maintained patches known as "isiduli." These are similar to some anthills that are frequently plowed under (Cadet et al., 2004). In a manner similar to this, it has been reported that Zimbabwean farmers employ anthill dirt to increase soil fertility (Bellon et al., 1999; Nyamapfene, 1986). The vast majority of studies have demonstrated that anthills contain much more total nitrogen (N) and exchangeable cations than neighboring soils (Holt and Lepage, 2000; Chikuvire et al., 2007). In tropical wet-dry climates, downslope erosion enhances soil fertility closer to anthills more than leached soils farther away, according to reports (Malaisse, 1978).

### Effect of Anthill Soil on Crops

Anthill soil is a practice that has been documented by farmers, and the bulk of these actions are supported by science (Watson, 1977; Nyamapfene, 1986). Utilizing "isiduli" somewhere in the field, according to some research, boosts sugar cane production by five times (Cadet et al., 2004). Anthills have been seen to have significantly higher plant biomass and grass growth than open regions in the Eastern Cape of South Africa (Steinke and Nell, 1989). According to study, an increase in grass growth is brought on by a buildup of runoff water at the base of anthills. This increase in grass growth boosts productivity during dry seasons and equips plants to endure even the most severe drought conditions. For instance, it was shown that *Cubitermes* soil in Burkina Faso, a country in West Africa, effectively managed the weed *Striga* infestation on sorghum crops (Andrianjaka et al., 2007).

Nyamangara and Nyagumbo (2010) investigated the interactive effects of selected nutrient resources and tied-ridging on plant growth performance in a semiarid smallholder farming environment in central Zimbabwe. They found that anthill soil and leaf litter are worthwhile investments for financially strapped farmers as they could improve the chemical and possibly physical properties of the soil. The possible hydrogen (pH) moderation effect may be advantageous to farmers who utilize them, ensuring the availability of nutrients like phosphorus that are often locked up in acidic soils.

According to Nyamangara et al. (2001), anthill soil and other organic fertilizers are better at protecting soils from acidification than mineral fertilizers. In an FYF (2011) study, it was found that some farmers in northern Malawi decided to take the soil from the anthill and mix it with goat manure in a ratio of one to one before applying it, improving the anthill soil's properties for plant nutrient content and also reducing the amount of manure required. This concept was thought to have a high likelihood of being adopted by other farmers who lack access to resources because it offers a cheap alternative to inorganic fertilizers. Additionally, substantial harvests were reported without any formal scientific study. According to FYF, smallholder farmers have also been observed in Malawi's central region spreading goat dung, compost, and anthill soil on their fields (2011). This implies a practical, affordable input that could counteract the consequences of high inorganic fertilizer prices.

Anthill soil can provide yields of more than 1000 kg/ha when applied appropriately in agricultural fields, according to study conducted by ZARI (2014) in Zambia. To learn how the technology influences crop response to medium-to-high rainfall circumstances, application rates, and soil management strategies, more research is required. Nevertheless, despite scientists' claims that anthill soil might replace inorganic fertilizers, some farmers do not level anthills due to restrictions (Logan et al., 1990). The reasons why farmers do not support leveling anthills to maximize land use and permit mechanized tillage operations have also been highlighted by scientists (Nyamapfene, 1986). These issues are said to disregard the financial (Nkunika, 1998) and spiritual (Geissler, 2000) importance that farmers attach to anthills.

Brossard et al. (2007) find that excessive exploitation of anthill soil can effect termite numbers in addition to nutrient mining. In addition, some farmers have asked assistance with the method, especially when digging (ZARI, 2014). Additionally, it appears that acquiring the proper quantities required to meet the nutritional requirements of crops has frequently been a problem when employing anthill soil in agricultural operations. Commercial farmers and growers of yearly seasonal crops are unable to profit from the anthills' low rates of production, claim Lee

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and Wood (1971). If scientists are aware of the drawbacks of using anthill soils for crop production, they will be able to come up with remedies and other ways to speed up the growth of anthills for agricultural output.

**(13) Methodology**(Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

### Study Areas and Coverage

The study will be conducted at the nursery of Capiz State University-Pontevedra.

### Experimental Methods

#### Planting Materials

Anthill soils will be gathered on and around the Capiz State University-Pontevedra Campus property. Pechay, lettuce, and mustard seeds will be purchased from KNOWN YOU SEED for use in this experiment. Only empty anthill mounds will be harvested. The mounds will be packed up and brought to the experiment area where they will be pounded into a powder, utilized as soil samples for the experiment, and crushed. On the other hand, the materials required to create a conventional soil medium, which consists of garden soil, sand, and cow dung, will be obtained locally. A total of 450 pots will be used in this pot experiment, broken down into 10 pots in each of the five soil treatment combinations, with three vegetable types replicated three times. The pots will be filled with well-ground and crushed anthill soil in combination with the standard soil medium as indicated in the various treatments. On the other hand, for the germination test of the various vegetable seeds, a total of 45 seed trays will be needed, which will be filled with various soil treatment combinations as indicated in the treatment.

#### Experimental Setup and Treatment

To start the pechay, lettuce, and mustard seeds for the germination trials, five (5) trays will be utilized for each vegetable, with various soil treatments in each tray. The five separate treatments for anthill soil and standard soil medium are as follows: T1 - 100 percent standard soil medium (SSM), T2 - 75 percent SSM + 25 percent anthill soil (AHS), T3 - 50 percent SSM + 50 percent AHS, T4 - 25 percent SSM + 75 percent AHS, and T5 - 100 percent AHS. With the help of the polyethylene plastic bags, the quantity required for each treatment for the combined soil treatments will be calculated based on a pot with a capacity of 2 kg. For the proper germination and growth experiment, the type of soil to be used will be thoroughly blended (homogenized).

A standard seed-germination experiment arranged in a completely randomized design (30 seeds per treatment) and replicated three times will be carried out in the nursery. Once the seeds are sown, the soil treatments will be moistened regularly (watering done twice a day (morning and evening)) and the number of germinated seeds will be counted and recorded for each treatment.

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Upon the completion of the germination experiment, the germinated seedlings will be transplanted into pots containing the respective five soil treatments. The pots will be arranged on top of tables using a completely randomized design (CRD). Water will be applied to the potted seedlings until they reach their field capacity moisture level, followed by incubation for a period of 3 days. Thereafter, the seedlings will be watered manually every twenty-four hours, and it will be done twice a day (early morning and late evening).

#### Physicochemical Properties on Anthill Soil and Standard Soil Medium

Due to limited resources, only the following parameters will be determined for the physicochemical analysis of the treatments to be used in this study: soil pH and electrical conductivity, soil texture, organic matter (percent), total nitrogen (percent), total phosphorus (percent), total potassium (percent), and carbon. This analysis will be carried out at the Soils Testing Laboratory of Capiz State University-Pontevedra Campus.

#### Data Collection

The quantity of seeds that germinate each day will be taken into consideration when observing the sown vegetable seeds, mostly for the purpose of data collecting. The number of seeds that germinated per treatment will be counted and documented three (3) days after nursing. According to Olisa et al. (2010), germination data will be used to calculate germination indices such as germination percentage (G percent), germination index (GI), and germination rate index (GRI).

#### Germination Percentage (G percent)

$$G\% = \frac{\text{Number of emerged seedlings at the final count}}{\text{Total Number of Seeds Planted}} \times 100$$

#### Germination Index (GI)

$$GI = \frac{\sum (N_x)(DAS)}{\text{Total number of seedlings that emerged at the final count}}$$

#### Germination Rate Index (GRI)

$$GRI = \frac{GI}{G\%}$$

#### Growth and Yield Data Collection

The following growth parameters will be measured: plant height, leaf length and width, and number of leaves. For each treatment, ten (10) plants of each tested crop will be harvested per treatment per replication at 25 days after transplanting to determine the fresh and dry weights of the whole plant and root biomass. The weights will be determined using a digital weighing scale. To determine dry weights, plant organs will be oven-dried at 60 °C for a period of two days and then weighed.

#### Postharvest Quality

The freshly harvested vegetables such as pechay, lettuce, and mustard will be sent to the nutritional laboratory in the area to determine their calories, total fat, total carbohydrates, dietary fiber, sugars, proteins, vitamins, and minerals.

#### Statistical Tools and Analysis

Collected data will be subjected to appropriate statistical tests.

**(14) Major Activities/Work plan**(Enumerate in chronological order the task to be undertaken. Use gantt chart. Below is an example. This can be modified base on your desired Work plan.)

| Activities                                        | 2022 |          |         |          |
|---------------------------------------------------|------|----------|---------|----------|
|                                                   | Nov  | December | January | February |
| Proposal Writing and Presentation                 |      |          |         |          |
| Area Preparation and Procurement of Needed Inputs |      |          |         |          |
| Collection and Preparation of Soil Media          |      |          |         |          |
| Physico-chemical Analysis of Soil Media           |      |          |         |          |

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| Conduct of Experiment                                                  |  |  |  |  |
| Germination Indices                                                    |  |  |  |  |
| Growth and Yield                                                       |  |  |  |  |
| Data Collection                                                        |  |  |  |  |
| Data on Germination Indices                                            |  |  |  |  |
| Data on Growth and Yield                                               |  |  |  |  |
| Postharvest Quality Assessment                                         |  |  |  |  |
| Writing of the Results and Discussion, Conclusions and Recommendations |  |  |  |  |
| Submission, Revision and Dissemination of Research Output              |  |  |  |  |
| Publication of Research Output in Refereed Journal and or Patenting    |  |  |  |  |

**(15) Expected Output**(Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

Publication of a Technoguide in the Production of Anthill-Grown Vegetables in Pots

**(15) Technology Transfer Activity Design**(Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Production of Premix Soil Media with Anthill Soil as Ingredient for Distribution and Use of Extension Beneficiaries

**(16) Targets Beneficiaries**(Who the clienteles are and what are the expected outcome/effects of the use of the projects output)

The study will help smallholder farmers who lack resources to produce organic fertilizer for their crops at the lowest possible cost using naturally occurring and plentiful resources found in any agri-ecological zone in the community. When the university decides to invest in the production of this soil amendment or fertilizer material, the research may also be advantageous to the institution as a whole.

**(17) Personnel Requirement**

| Name                     | Role<br>(Program,Project, Study Leader) | Field of Specialization       |
|--------------------------|-----------------------------------------|-------------------------------|
| Dr. Diego M. Malones     | Project Leader                          | Agronomy                      |
| Mr. Jon Verny D. Biaco   | Co-Project Leader                       | Soil Science/Agronomy         |
| Dr. Eden A. Alam         | Co-Project Leader                       | Agronomy                      |
| Mr. Junarey B. Doletin   | Co-Project Leader                       | Agricultural Economics        |
| Mr. Jasper A. Basalo     | Co-Project Leader                       | Horticulture                  |
| Dr. Harold O. Buenvenida | Co-Project Leader                       | Biology/Environmental Science |
| Mr. Darel D. Dapulaza    | Co-Project Leader                       | Agronomy                      |
| Mr. Martin B. Ton-ogan   | Co-Project Leader                       | Agronomy                      |

**Percent time devoted to the Program/ Project/ Study**

**Program/Project Leader (in hours):**

**Program/Project Members (in hours):**

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### (18) Budgetary Requirement

| SOURCES OF EXPENSES                                                                                         | Quantity | Amount (PhP)         |
|-------------------------------------------------------------------------------------------------------------|----------|----------------------|
| <b>A. Personnel Services</b> (For studies needed the Research Assistant, informant/s, guide)                |          |                      |
| <b>B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)</b> |          |                      |
| -Collection and Preparation of Soil Media                                                                   |          | 2,000.00             |
| -Procurement of Polybags and seedtrays                                                                      |          | 5,000.00             |
| -Physico-Chemical Analysis of Soil Media                                                                    |          | 1,500.00             |
| -Postharvest Quality Analysis                                                                               |          | 20,000.00            |
| -Materials and Supplies                                                                                     |          | 3,000.00             |
|                                                                                                             |          |                      |
| <b>C. CAPITAL OUTLAY</b>                                                                                    |          |                      |
| <b>Total</b>                                                                                                |          | <b>PhP 31,500.00</b> |

### (19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

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